Rajdeep Gill

431-998-9785 | gillr10@myumanitoba.ca | linkedin.com/in/rajdeep-gill | github.com/Rajdeep-Gill

Education

University of Manitoba — Winnipeg, MB
Bachelor of Science in Computer Engineering — *Sept. 2021 - Present*

- GPA: *4.45/4.5*
- Relavent Coursework: *Engineering Algorithms, Data structures and Algorithms, Systems Engineering Principles*

Experience

Undergraduate Research Student — May 2024 - Current
University of Manitoba — *Winnipeg, MB*

- Implemented a centroid detection algorithm using computer vision techniques to identify the center of cells in optical data, achieving a success rate on 92% of cells
- Designed and trained a neural network model using TensorFlow and Keras to predict cell viability from optical data, resulting in an 85% accuracy rate
- Developed a tKinter-based GUI for data input and cell visualization, enhancing user experience with an interactive interface for analysis and data observation

Undergraduate Research Student — May 2023 - August 2023
University of Manitoba — *Winnipeg, MB*

- Employed dimensionality reduction techniques, including t-SNE (t-distributed Stochastic Neighbor Embedding) and PCA (Principal Component Analysis), to compare and cluster data
- Implemented parallel processing to accelerate the gathering and processing of data, achieving a two-fold reduction in runtime

Projects

Discord Clone | *React, TypeScript, HTML/CSS, Next.js*

- Developed a feature-rich Discord Clone using Next.js, React, and TypeScript, providing users with a seamless and familiar real-time communication experience
- Utilized Tailwind CSS to enhance the front-end design, ensuring a visually appealing and responsive user interface on both desktop and mobile
- Leveraged PlanetScale as the database solution for storing messages, servers, and user accounts
- Implemented authentication using Clerk for secure user login and registration via Email or Google
- Integrated LiveKit for the implementation of high-quality video and voice calls, and web sockets for real-time messaging
- Provided users with extensive control, allowing them to edit and delete messages, create and delete servers, and manage server members effectively

Chess Game | *Python*

- Created a command line chess game using Python
- Implemented features such as move validation, checkmate detection, and piece promotion
- Currently developing multiplayer functionality using socket programming to enable real-time play between users

Awards

NSERC Undergraduate Student Research Award — 2023, 2024
Presidents Scholarship — 2021-2024

Technical Skills

Languages: Python, Java, C, JavaScript, HTML/CSS, R
Frameworks: React
Developer Tools: Git, VS Code, Visual Studio, Android Studio, IntelliJ
Libraries: Pandas, NumPy, Matplotlib, tKinter
Other: Microsoft Office Suite